

# **RRUHFR03**

## **Parking Mode User Manual v1.0**



**RapidRadio Solutions Private Limited**

B-404, Satyamev 1, Opp. Gujarat High Court,  
S. G. Highway, Ahmedabad - 380060, Gujarat, India

[www.rapidradio.co.in](http://www.rapidradio.co.in) [support@rapidradio.co.in](mailto:support@rapidradio.co.in) +91 79 2766 5256



## Index

| Sr. No. | Topic                                 | Page |
|---------|---------------------------------------|------|
| 1       | About the Document                    | 3    |
| 2       | Tools Used                            | 4    |
| 3       | Understanding Parking Mode in the SDK | 5    |
|         | Tag Registration                      |      |
|         | General Statistics                    |      |
|         | Tag entry modification                |      |
|         | Category-Relay(s) Selection           |      |
|         | Relay(s) Configuration                |      |
|         | TID Persistence                       |      |
|         | Administrative Configuration(s)       |      |
|         | Storage Configuration                 |      |
|         | Get Logged TID count                  |      |
|         | Start Log read                        |      |
| 4       | Basic Communication Frame Format      | 10   |
| 5       | Command Frames                        | 12   |
| 6       | Revision History                      | 37   |



## **About the Document**

This document is applicable for RapidRadio UHF Integrated RFID Reader family product, i.e. RRUHFR03 with built-in memory. It is intended to explain various functionality available in the SDK using detailed description along with application screenshots.

If you are looking for the User Manual or Communication Protocol Document, this is not the correct document. Please write back your actual requirements to us so we can provide you with the correct documentation.

After comparing this document with the application UI, a thought may arise that the document does not describe some of the functionalities available in the application. Please note that such configurations / operations are either not suitable or deprecated for new development and so the same have been omitted.



## **Tools Used**

The SDK actually contains a Windows Forms Application based on .Net Framework 4.8, which has been coded in C#.Net. It uses bare metal code for demonstrating the RFID reader functionality, ensuring complete control of code to the system integrator. Only one external DLL named ZedGraph.dll is used to represent the RF noise data as a graph.

If you need the SDK for any other platform, feel free to write back to us. We may provide a basic SDK for the desired platform in case it is readily available with us. And even if it is not available, we can still help you integrate the RFID readers with your software.



## Understanding Parking Mode in the SDK

The screenshot displays the 'RRUHFOEM07 Test App. v1.9a' window with the 'Parking mode' tab selected. The interface is divided into several configuration sections:

- Reader Config:** Includes tabs for EPC C1G2 Operations, Extended Inventory Config, Parking mode (selected), RF diagnosis, TCP device search, TCP Server, and Fault Diagnosis.
- Communication setup:**
  - COM Port:** Radio button selected. Fields for COM port (dropdown), Baudrate (115200), and a 'Connect' button.
  - TCP/IP:** Radio button unselected. Fields for Device IP (192.168.1.32), TCP Port (6000), and radio buttons for TCP Server and TCP Client.
- Device information and basic configurations:**
  - Device information:** Fields for Device SN, FW Version, and HW Version, with a 'Get' button.
  - Device operating mode setup:** 'Device working mode' dropdown set to 'Response mode', with 'Get' and 'Set' buttons.
  - RF info:** A text area for RF information.
  - Global Password:** A text field with 'Authenticate', 'DeAuthenticate', and 'Update' buttons.
  - Heartbeat Setup:** 'x100mS' field with a 'Setup' button and a 'TCP Client check' checkbox.
  - Device RTC setup:** 'Internal RTC time' field with 'Get' and 'Synchronize' buttons.
  - Device restart and control:** 'Device restart' and 'Force Boot mode' buttons.
  - Buzzer control:** 'Buzzer enable' checkbox, 'Buzzer setup' button, 'Buzzer Beep' text field, and 'Buzzer beep (x100mS)' button.
- RF Configuration:**
  - Region: INDIA (dropdown), Link Mode: 1 (dropdown), Min. Frequency, Max. Frequency, and RF Power fields, each with 'Get' and 'Set' buttons.
- TCP/IP configuration:**
  - Device IP, Gateway, Net mask, and MAC fields, each with a checkbox and 'Get'/'Set' buttons.
  - Device Port, Host Port, and Host IP fields, each with a checkbox and 'Get'/'Set' buttons.
  - DHCP checkbox with 'Get' and 'Set' buttons.
- Device UART setup:** 'Device Baudrate' dropdown set to 921600, with 'Get' and 'Set' buttons.
- Relay Control (Oneshot):** Four rows for Relay-1 to Relay-4, each with a value field (10) and a 'Set' button.
- Relay Control(Fixed):** Four checkboxes for Relay-1 to Relay-4, with a 'Set' button.
- Mux. Antenna Port:** 'Multiplexer port' dropdown set to 1, with 'Get' and 'Set' buttons.

This operating mode will be implemented in the RRUHFR03 Reader, which is integrated with onboard memory for data storage. In this mode, the reader is designed to automatically detect and read the TID (Tag Identifier) of any RFID tag or card presented to it. Upon detection, both the associated data and operational logs such as records of tag entries will be directly stored within the reader's internal memory.

This allows the device to function independently without relying on an external system or continuous network connection, making it suitable for standalone applications where reliable tag identification and data logging are essential.



## Tag Registration

Tag registration is the process of registering tag/card information in the system. There are two methods available for registering a tag:

### Method 1

Need to update the device working mode to "Response mode" and place the tag/card within the reader's range, ensuring it remains steady and unmoved. Click Query, and once the TID (Tag Identifier) is displayed, select the appropriate category, check the Whitelist Enable box, and click Register after filling in all required details.

### Method 2

If you already have a list of TIDs, you can manually enter each TID into the system in the "Parking mode" section. Then, select the corresponding category, tick the Whitelist Enable checkbox, and click Register once all necessary details have been provided.

A screenshot of the 'Tag registration' dialog box. It contains a 'TID:' label followed by a text input field. Below this is a 'Category:' label with four radio button options labeled 1, 2, 3, and 4. Option 1 is selected. Below the category options is a checkbox labeled 'Whitelist Enable' which is checked. At the bottom of the dialog are two buttons: 'Query' and 'Register'.

## General Statistics

When you click the "Check Tag Entries" button, the total number of registered Whitelisted and Blacklisted tag entries will be displayed in their respective fields.

When you click on the "Start Record(s) Read" button, the list of registered EPC tags becomes available on the EPC C1G2 Operations screen.

A screenshot of the 'General statistics' dialog box. It contains two labels: 'Whitelisted entries:' and 'Blacklisted entries:', each followed by a text input field. Below these fields are two buttons: 'Check tag entries' and 'Start Record(s) Read'.



## Tag entry modification

If any modifications are required in the Category or Status of registered tags/cards, you can view, update, or delete existing entries from the records as needed.

The 'Tag entry modification' form contains the following fields and controls:

- TID:** A text input field for the Tag ID.
- Category:** Four radio buttons labeled 1, 2, 3, and 4.
- Status:** Two radio buttons labeled 'Whitelist' and 'Blacklist'.
- Buttons:** Three buttons at the bottom labeled 'Check', 'Update', and 'Delete'.

**Whitelist** indicates that the tag is Authorized/Approved.

**Blacklist** indicates that the particular Tag is Blocked/Unapproved.

## Category-Relay(s) selection

The relay can be selected for each specific category and assigned to the corresponding tag. When a tag is detected, the relay associated with its category will be activated.

The system includes four relays in total—two relays are available as “Standard”, while the remaining two are “Optional”. Based on the application requirements, each category can be assigned to a particular relay.

The 'Category-Relay(s) selection' form contains the following fields and controls:

| Category   | R1                       | R2                       | R3                       | R4                       |
|------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Category-1 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Category-2 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Category-3 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Category-4 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

At the bottom of the form are two buttons labeled 'Get' and 'Set'.



## Relay(s) configuration

This command can be helpful in setting the desired relay for a specific time interval. The textboxes accept a 16 bit unsigned integer value which will be multiplied by the reader with 100 in order to derive total milliseconds for which the relay will stay active. After that time interval, it will be automatically switched off.

The 'Relay(s) Configuration' dialog box contains a section titled 'On Time'. It lists four relays, each with a text input field followed by 'x 100mS'. Below the inputs are two buttons: 'Get' and 'Set'.

| Relay    | On Time (x 100mS)    |
|----------|----------------------|
| Relay 1: | <input type="text"/> |
| Relay 2: | <input type="text"/> |
| Relay 3: | <input type="text"/> |
| Relay 4: | <input type="text"/> |

Buttons: Get, Set

## TID Persistence

The persistence feature is used to prevent the RFID reader from repeatedly sending the same tag's data within a specified time interval. When persistence is enabled, the reader measures the time between two consecutive detections of the same tag and compares it to the configured persistence time. If the elapsed time exceeds the persistence time, the tag data is transmitted to the remote server or host; otherwise, the duplicate detection is ignored.

The auto reset feature allows the reader to automatically resend data for the same tag when it remains steady within the reader's field. Once the tag/card is detected and held steady, the reader will continue to detect and transmit the tag data at regular intervals without requiring the tag to be removed from the field.

The 'TID Persistence' dialog box includes a 'Time (x100ms):' label and a text input field containing the value '10'. Below this is an unchecked checkbox labeled 'Auto reset'. At the bottom are 'Get' and 'Set' buttons.

Time (x100ms):

☐ Auto reset

Buttons: Get, Set

When enabled, the **Log device SN** will include the 4 byte reader serial number in the tag data packet.

The 'Log device SN' dialog box features an unchecked checkbox labeled 'Log device SN'. Below the checkbox are 'Get' and 'Set' buttons.

☐ Log device SN

Buttons: Get, Set





### Administrative configuration(s)

The log records both whitelisted, blacklisted and unregistered tags/cards by default. If the "Log registered only" option is selected, the log will include only the whitelisted and blacklisted tags/cards.

Administrative configuration(s)

Offline TID logging

☐ Log registered only ☐ Log all

Get Set

### Storage configuration

To clear the detected tags/cards log, use the "**Log erase**" option.

To delete all registered tags/cards and the log of detected tags/cards, select the '**Erase all**' option.

Storage configuration

Log erase

Erase all

The **Get Logged TID Count** button is used to display the number of detected tags/cards in the designated display area.

Get Logged TID count

Start Log Read

The **Start log read** button is used to read the log of detected tags/cards, the data will be displayed in the EPC C1G2 Operations section when TCP Client is selected in the Log data routing settings.

If **TCP Server** is selected, the log of detected tags/cards will be displayed in the **TCP Server** section after setting the **listen port** and clicking **Start**.

Log data routing

☐ TCP Server ☐ TCP Client

Get Set

Listen Port 9000 Start



## Basic Communication Frame Format

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data       |
|---------|-------------------------------|-------------------|------------|
| 1.      | SOF                           | 1                 | BBHex      |
| 2.      | Request Frame Length          | 1                 | XXHex      |
| 3.      | Command Code                  | 1                 | XXHex      |
| 4.      | Sub Command                   | 1                 | 80Hex      |
| 5.      | Custom Request Data           | X                 | XX...XXHex |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex    |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data       |
|---------|-------------------------------|-------------------|------------|
| 1.      | SOF                           | 1                 | BBHex      |
| 2.      | Response Frame Length         | 1                 | XXHex      |
| 3.      | Command Code                  | 1                 | XXHex      |
| 4.      | Error Code                    | 1                 | XXHex      |
| 5.      | Sub Command                   | 1                 | 80Hex      |
| 6.      | Custom Response Data          | X                 | XX...XXHex |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex    |

### WHERE

**SOF :** SOF is known as Start of frame. The byte BBHex should be transmitted in the starting of the frame.

**Request / Response Frame Length:** This field helps identify total bytes being transferred between the module and the host. For a request, all the bytes starting from "Command Code" to "Custom Request Data" field are included in calculation of Request Frame Length. Similarly, Response Frame Length has the count of all bytes between "Command Code" and "Custom Response Data", including both these fields.

**Command Code:** One byte command code is used to identify the transaction that is required to be performed by the module.

**Sub Command Code:** One byte command code is used to identify the type of process to be done which is required to be performed by the module. The command code is always transmitted in MSByte first form.



**Error Code:** Error code is included in all response frames and are used to identify the exact cause of a failed transaction. Two main values of Error Code are 00Hex = Success (No Error) and F0Hex = Generic Error.

**Custom Request / Response Data:** This field contains data that is related to the specific request or response being transferred between the module and the host. In case any specific field spans over multiple bytes, the same shall be always transmitted in MSByte first form.

**Cyclic Redundancy Check (CRC):** CRC field should be used to validate the data frame, either a request or a response. It ensures that the data being transferred between the module and the host is intact and valid. The value of CRC is calculated over the data starting from the "Command Code" field up to "Custom Request/Response Data" field. Being a 2 byte field, it will always be transmitted in MSByte first form.

The logic to calculate the CRC value is given below.

```
unsigned int CalculateCRC()
{
    unsigned int crc = 0x0000FFFF; //32 bit unsigned int
    unsigned char bytes, bits; // 8 bit unsigned int

    for (bytes = 2; bytes < txBuffer[1] + 2; bytes++)
    {
        crc = crc ^ txBuffer[bytes];
        for (bits = 0; bits < 8; bits++)
        {
            if ((crc & 0x8000) == 0x8000)
                crc = (crc << 1) ^ 0x1021;
            else
                crc = (crc << 1);
        }
    }
    crc = (~crc);
}
```

Following is a usage example of the CalculateCRC method.

```
void main()
{
    unsigned char txBuf[256]; // 8 bit unsigned int
    unsigned int crc;

    txBuf[0] = 0xBB; // SOF
    txBuf[1] = 0x01; // Request Length
    txBuf[2] = 0x01; // Command Code
    crc = CalculateCRC();
    txBuf[3] = (unsigned char)((crc >> 8) & 0xFF);
    txBuf[4] = (unsigned char)(crc & 0xFF);

    // Request frame is ready
    // Same CRC calculation logic applies to response frame also
}
```



## Command Frames

### Query The Tag

This packet is used to get the TID of the tag in the device.

#### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | 21Hex   |
| 4.      | Sub Command                   | 1                 | 80Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

#### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | 21Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Tag Count                     | 1                 | XXHex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

#### TID Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data       |
|---------|-------------------------------|-------------------|------------|
| 1.      | SOF                           | 1                 | BBHex      |
| 2.      | Response Frame Length         | 1                 | 0FHex      |
| 3.      | Command Code                  | 1                 | E0Hex      |
| 4.      | Sub Command Code              | 1                 | 02Hex      |
| 5.      | TID Length                    | 1                 | 0CHex      |
| 6.      | TID                           | 12                | XX...XXHex |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex    |



## **WHERE**

**TID Length:** Length of the TID.

**TID Data:** TID Data assigned with the tag.

**Error Code:** Possible Values : 00Hex = Success.  
B1Hex = No tag found.



## Register The Tag

This packet is used to register the tag in the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | XXHex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 01Hex   |
| 5.      | Category                      | 1                 | XXHex   |
| 6.      | RFU                           | 1                 | XXHex   |
| 7.      | Tag Status                    | 1                 | XXHex   |
| 8.      | TID Length                    | 1                 | XXHex   |
| 9.      | TID Data                      | XX                | XXHex   |
| 10.     | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 01Hex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Category:** Category of selected Relays in the TID entry.

**Tag Status:** Possible Values : 02Hex = Whitelisted data.  
04Hex = Blacklisted data.

**TID Length:** Length of the TID.

**TID Data:** TID Data assigned with the tag.

**Error Code:** Possible Values : 00Hex = Success.  
06Hex = Tag already registered.



## Check Tag Entry

This packet is used to check the count of the registered tag entries of the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 04Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 07Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 04Hex   |
| 6.      | Whitelisted Entries           | 2                 | XXXXHex |
| 7.      | Blacklisted Entries           | 2                 | XXXXHex |
| 8.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Whitelisted Entries:** It indicates the number of tags are Authorized/approved.

**Blacklisted Entries:** It indicates the number of tags are Blocked/unapproved.



## Read Record

This packet is used to read all the records of the registered tag.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 05Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | XXHex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 05Hex   |
| 6.      | RFU                           | 1                 | XXHex   |
| 7.      | Tag Status                    | 1                 | XXHex   |
| 8.      | Category                      | 1                 | XXHex   |
| 9.      | TID Length                    | 1                 | XXHex   |
| 10.     | TID Data                      | XX                | XXHex   |
| 11.     | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Tag Status:** Possible Values : 02Hex = Whitelisted data.  
04Hex = Blacklisted data.

**Category:** Category of selected Relays in the TID entry.

**TID Length:** Length of the TID.

**TID Data:** TID Data assigned with the tag.





## Tag Entry Modification Check

This packet is used to check the entry of the tag in the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 0FHex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 09Hex   |
| 5.      | TID Length                    | 1                 | XXHex   |
| 6.      | TID Data                      | XX                | XXHex   |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | XXHex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 09Hex   |
| 6.      | Category                      | 1                 | XXHex   |
| 7.      | RFU                           | 1                 | XXHex   |
| 8.      | Tag Status                    | 1                 | XXHex   |
| 9.      | TID Length                    | 1                 | XXHex   |
| 10.     | TID Data                      | XX                | XXHex   |
| 11.     | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Category:** Category of selected Relays in the TID entry.

**Tag Status:** Possible Values : 02Hex = Whitelisted data.  
04Hex = Blacklisted data.

**TID Length:** Length of the TID.

**TID Data:** TID Data assigned with the tag.



## Update Tag Entry

This packet is used to update the entry of the tag in the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 12Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 0AHex   |
| 5.      | Category                      | 1                 | XXHex   |
| 6.      | RFU                           | 1                 | XXHex   |
| 7.      | Tag Status                    | 1                 | XXHex   |
| 8.      | TID Length                    | 1                 | XXHex   |
| 9.      | TID Data                      | XX                | XXHex   |
| 10.     | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 0AHex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Tag Status:** Possible Values : 02Hex = Whitelisted data.  
04Hex = Blacklisted data.

**Category:** Category of selected Relays in the TID entry.

**TID Length:** Length of the TID.

**TID Data:** TID Data assigned with the tag.

**Error Code:** Possible Values : 00Hex = Success.  
FFHex = Error.



## Delete Tag Entry

This packet is used to delete the entry of the tag in the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 0FHex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 0BHex   |
| 5.      | TID Length                    | 1                 | XXHex   |
| 6.      | TID Data                      | XX                | XXHex   |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 0BHex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**TID Length:** Length of the TID.

**TID Data:** TID Data assigned with the tag.



## Get Relay Category Configuration

This packet is used to get the configurations of the relay categories assigned with the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 04Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 10Hex   |
| 5.      | Category Start Address        | 1                 | XXHex   |
| 6.      | Category End Address          | 1                 | XXHex   |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | XXHex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 10Hex   |
| 6.      | Category X Configuration      | X                 | XXHex   |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

#### Category Configuration:

| Category No. | Bit 7 (MSBit) | Bit 6 | Bit 5 | Bit 4 | Bit 3   | Bit 2  | Bit 1   | Bit 0 (LSBit) |
|--------------|---------------|-------|-------|-------|---------|--------|---------|---------------|
| Category X   | RFU           | RFU   | RFU   | RFU   | Relay 4 | Relay3 | Relay 2 | Relay 1       |



## Set Relay Category Configuration

This packet is used to set the configurations of the relay categories.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data       |
|---------|-------------------------------|-------------------|------------|
| 1.      | SOF                           | 1                 | BBHex      |
| 2.      | Request Frame Length          | 1                 | XXHex      |
| 3.      | Command Code                  | 1                 | A0Hex      |
| 4.      | Sub Command                   | 1                 | 11Hex      |
| 5.      | Category Start Address        | 1                 | XXHex      |
| 6.      | Category End Address          | 1                 | XXHex      |
| 7.      | Category Configuration        | X                 | XX...XXHex |
| 8.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex    |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 11Hex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

#### Category Configuration:

|            | Bit 7 (MSBit) | Bit 6 | Bit 5 | Bit 4 | Bit 3   | Bit 2  | Bit 1   | Bit 0 (LSBit) |
|------------|---------------|-------|-------|-------|---------|--------|---------|---------------|
| Category X | RFU           | RFU   | RFU   | RFU   | Relay 4 | Relay3 | Relay 2 | Relay 1       |



## Get Relay Configurations

This packet is used to get the configurations about the relays of the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 04Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 12Hex   |
| 5.      | Relay Start Address           | 1                 | XXHex   |
| 6.      | Relay End Address             | 1                 | XXHex   |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | XXHex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 12Hex   |
| 6.      | Duration of Relay             | x                 | XXXXHex |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Duration of Relay:** The duration of the Relay will be in multiples of 100 milliseconds.



## Set Relay Configurations

This packet is used to set the configurations about the relays of the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 04Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 13Hex   |
| 5.      | Relay Start Address           | 1                 | XXHex   |
| 6.      | Relay End Address             | 1                 | XXHex   |
| 7.      | Duration of Relay             | X                 | XXXXHex |
| 8.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 13Hex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

## WHERE

**Duration of Relay:** The duration of the Relay will be in multiples of 100 milliseconds.



### Get TID Persistence Time

This packet is used to get the parameters about the TID Persistence time and auto reset flag of the device.

#### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 20Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | 771AHex |

#### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 06Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 20Hex   |
| 6.      | Auto Reset Flag               | 1                 | XXHex   |
| 7.      | Duration for persistence      | 2                 | XXXXHex |
| 8.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

#### WHERE

**Auto Reset Flag:** The flag which decides that the reader should automatically resend data for the same tag or not when it remains steady within the reader's field.

Possible Values : 00Hex = Reader will not send the data for the same tag.

01Hex = Reader will send the data for the same tag.

**Duration for persistence:** The duration of the persistence will be in multiples of 100 milliseconds.





## Set TID Persistence

This packet is used to set the parameters about the TID Persistence time and auto reset flag of the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 05Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 21Hex   |
| 5.      | Auto Reset Flag               | 1                 | XXHex   |
| 6.      | Persistence Time              | 2                 | XXXXHex |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 21Hex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

## WHERE

**Auto Reset Flag:** The flag which decides that the reader should automatically resend data for the same tag or not when it remains steady within the reader's field.

Possible Values : 00Hex = Reader will not send the data for the same tag.

01Hex = Reader will send the data for the same tag.

**Duration for persistence:** The duration of the persistence will be in multiples of 100 milliseconds.



## Get Data Routing Configuration

This packet is used to get the parameters to transmit response in TCP server or TCP client.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 22Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | 751AHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 04Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 22Hex   |
| 6.      | Data Routing Type             | 1                 | XXHex   |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Data Routing Type:** The flag which decides that the reader should transmit the data in TCP Server or in TCP client.

Possible Values : 02Hex = TCP Server.

04Hex = TCP Client.



## Set Data Routing Configuration

This packet is used to set the parameters to transmit response in TCP server or TCP client.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 23Hex   |
| 5.      | Data Routing Type             | 1                 | XXHex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 23Hex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Data Routing Type:** The flag which decides that the reader should transmit the data in TCP Server or in TCP client.

Possible Values : 02Hex = TCP Server.

04Hex = TCP Client.



## Get TID Offline Log Configuration

This packet is used to get the configuration of the TID offline log of the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 24Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | 731AHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 04Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 24Hex   |
| 6.      | Log Type                      | 1                 | XXHex   |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Log Type:** Possible Values : 01Hex = Only Registered Tag logs.  
02Hex = All logs.



## Set Offline TID Log Configuration

This packet is used to set the configuration of the offline TID log of the device.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 25Hex   |
| 5.      | Log Type                      | 1                 | XXHex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 25Hex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

### WHERE

**Log Type:** Possible Values : 01Hex = Only Registered Tag logs.  
02Hex = All logs.



## Get Logged TID Count

This packet is used to get the total log count of the tag entries.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 26Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | 711AHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data       |
|---------|-------------------------------|-------------------|------------|
| 1.      | SOF                           | 1                 | BBHex      |
| 2.      | Response Frame Length         | 1                 | 07Hex      |
| 3.      | Command Code                  | 1                 | A0Hex      |
| 4.      | Error Code                    | 1                 | XXHex      |
| 5.      | Sub Command Code              | 1                 | 26Hex      |
| 6.      | Total Count                   | 4                 | XX...XXHex |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex    |

### WHERE

**Total Count:** Total tag entries count in Hex.



## Read Log

This packet is used to read the log entry of the registered tag.

### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 27Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | 701AHex |

### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data       |
|---------|-------------------------------|-------------------|------------|
| 1.      | SOF                           | 1                 | BBHex      |
| 2.      | Response Frame Length         | 1                 | XXHex      |
| 3.      | Command Code                  | 1                 | E2Hex      |
| 4.      | Tag Status                    | 1                 | XXHex      |
| 5.      | Information Flag              | 1                 | XXHex      |
| 6.      | Device Serial Number          | 4                 | XX...XXHex |
| 7.      | RTC Data                      | 6                 | XX...XXHex |
| 8.      | TID Length                    | 1                 | XXHex      |
| 9.      | TID Data                      | XX                | XXHex      |
| 10.     | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex    |

### WHERE

**Tag Status:** Possible Values : 00Hex = Unregistered Tag data.

02Hex = Whitelisted Tag data.

04Hex = Blacklisted/Blocked Tag data.

### Information Flag:

| Bit 7 (MSBit) | Bit 6         | Bit 5 | Bit 4 | Bit 3 | Bit 2 | Bit 1 | Bit 0 (LSBit) |
|---------------|---------------|-------|-------|-------|-------|-------|---------------|
| RTC           | Serial Number | RFU   | RFU   | RFU   | RFU   | TID   | RFU           |



**Device Serial Number:** The serial number assigned to the device.

**RTC Data:** It is a Real-Time Clock data of 6 bytes in hex.

Format of RTC data: **HH MM SS MM DD YY** [H: Hour; M: Minute; S: Second; M: Month; D: Day; Y: Year] in Hex.

**TID Length:** Length of the TID.

**TID Data:** TID Data assigned with the tag.





### Get Device Serial Number Status in TID Response

This packet is used to get the status of the device serial number in the TID response frame is enabled or disabled .

#### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 28Hex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

#### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 04Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 28Hex   |
| 6.      | Serial number Flag            | 1                 | XXHex   |
| 7.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

#### WHERE

**Serial number Flag:** Possible Values : 00Hex = Enable.  
01Hex = Disable.



### Set Device Serial Number Status in TID Response

This packet is used to enable or disable the device serial number in the TID response frame.

#### Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | 29Hex   |
| 5.      | Serial number Flag            | 1                 | XXHex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

#### Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | 29Hex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

#### WHERE

**Serial number Flag:** Possible Values : 00Hex = Enable.  
01Hex = Disable.



## Log Erase

This packet is used to erase all the log entries of the registered as well as unregistered tags stored in the device.

## Request

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | FCHex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | AB1AHex |

## Response

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | FCHex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |

**Erase All Entries**

This packet is used to erase all the log entries as well as the registered tags which are stored in the device.

**Request**

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Request Frame Length          | 1                 | 02Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Sub Command                   | 1                 | FDHex   |
| 5.      | Cyclic Redundancy Check (CRC) | 2                 | AA1AHex |

**Response**

| Sr. No. | Parameter                     | Length (in Bytes) | Data    |
|---------|-------------------------------|-------------------|---------|
| 1.      | SOF                           | 1                 | BBHex   |
| 2.      | Response Frame Length         | 1                 | 03Hex   |
| 3.      | Command Code                  | 1                 | A0Hex   |
| 4.      | Error Code                    | 1                 | XXHex   |
| 5.      | Sub Command Code              | 1                 | FDHex   |
| 6.      | Cyclic Redundancy Check (CRC) | 2                 | XXXXHex |



### Revision History

| Sr. No. | Revision Date | Details              | Revised By | Approved By |
|---------|---------------|----------------------|------------|-------------|
| 1       | 25/06/2025    | Initial Draft - v1.0 | VM & KG    | DU          |